

**Xinmiao Huang¹, Qisong He¹, Zhenglin Huang¹, Boxuan Wang¹, Zhuoyun Li¹,
Guangliang Cheng¹, Yi Dong^{1,*}, Xiaowei Huang^{1*}**

a) Comparison of Existing Benchmarks

Existing Spatial Bench

CV-Bench
Q: Considering the relative positions of the person and the car in the image provided, where is the person located with respect to the car?
Extrinsic Static

BLINK
Q: Is the car under the cat?
Q: Which point is closer to the camera?
Extrinsic Static

What'sUp
A mug to the left of a cup
A mug in front of a cup
A mug behind a cup
A mug to the right of a cup
Extrinsic Static

Spatial-DISE
Q: Work out which of the cube can be made from the net.
Q: Work out which 3D figure in the grey box has been rotated to make the new 3D figure.
Intrinsic Dynamic

b) Framework of Spatial-DISE

Intrinsic-Dynamic:
- 2D/3D Folding
- 2D/3D Rotation
- Fold and Punch

Extrinsic-Dynamic:
- 2D Combination
- 3D Combination

Spatial-DISE
16% Intrinsic-Static
12% Extrinsic-Static
2D Shape Finding
3D Shape Finding
3D Projection

c) VLMs & Human Performances

d) Synthetic Data Generation Pipeline

blender Verifiable Generation

Initialization & Seeding → Core Asset Generation → Distraction Generation

Question and Correct Answer Rendering → Geometric Variation → Pattern/Orient. Errors → Incorrect Views → Component Replacement

Distraction Strategy

Controlled Rendering & Formatting → Large-scale Raw Synthetic Data → Duplication → Multi Answers → Missing

Quality Control

RealWorld Data + Synthetic Data

e) Statistics of Spatial-DISE Bench and Dataset

Spatial-DISE Bench

Task	Percentage
2D Rotation	4%
2D Folding	14%
2D Combination	4%
3D Combination	11%
3D Shape Finding	12%
3D Folding	12%
3D Rotation	13%
3D Projection	12%
Fold and Punch	14%

Spatial-DISE 12K

Task	Percentage
3D Shape Finding	19%
3D Combination	20%
3D Folding	21%
3D Projection	20%
3D Rotation	20%

ABSTRACT

*Corresponding author: xiaowei.huang@liverpool.ac.uk, yi.dong@liverpool.ac.uk

1 INTRODUCTION

Recent advances in vision language models (VLMs) have demonstrated impressive capabilities in various tasks such as object detection (Li et al., 2022; Peng et al., 2023; Anil et al., 2025), scene caption (Alayrac et al., 2022; Chen et al., 2023; Li et al., 2023), and visual question answering (Wang et al., 2023; Anil et al., 2025; Alayrac et al., 2022). However, their capability for sophisticated *dynamic* spatial reasoning, a cornerstone of human cognition and a critical requirement for applications in robotics, augmented reality, and autonomous navigation, remains a significant limitation (Ramakrishnan et al., 2024) and largely under-evaluated.

Existing benchmarks for evaluating the spatial reasoning of VLMs have three major limitations. Firstly, current benchmarks **lack a systematic cognitive framework** for categorizing and evaluating different types of spatial reasoning abilities, leading to fragmented, unbalanced tasks that typically focus only on basic skills (Liu et al., 2023; Chen et al., 2024; Cheng et al., 2024). Consequently, there is a notable scarcity of benchmarks designed to evaluate deeper cognitive abilities. Secondly, current benchmarks are **limited in scope**, focusing predominantly on *static* spatial questions that do not require *multi-step dynamic* reasoning (Wang et al., 2024; Han et al., 2020). Consequently, crucial cognitive abilities like mental rotation and folding are significantly under-tested. Thirdly, the few benchmarks that address dynamic tasks are **insufficient in scale** (Ray et al., 2024; Ramakrishnan et al., 2024), making them insufficient to robustly evaluate the capabilities of the model or to drive further model development.

To bridge these gaps, we propose **Spatial-DISE**. Unlike previous benchmarks that focus on isolated abilities or static scenes, Spatial-DISE adopts a unified 2x2 cognitive taxonomy (Maier, 1996; Uttal et al., 2013), as illustrated in Figure 1 (b), which covers both 2D and 3D aspects, and critically, places a strong emphasis on *dynamic* spatial reasoning tasks. The first dimension distinguishes between **intrinsic** information, which defines an object by its internal parts and their arrangement, and **extrinsic** information, which pertains to the spatial relations among different objects; the second dimension differentiates **static** tasks, which involve fixed and stationary information, from **dynamic** tasks, which require mental transformation. Figure 1 provides an overview of the Spatial-DISE framework, generation pipeline, and benchmark statistics.

Spatial-DISE contains more than 12K verified spatial reasoning Visual Question-Answer (VQA) pairs. It is created through a combination of real-world data collection and synthetic generation using Blender¹. Firstly, it has a set of 559 real-world and synthetic VQA pairs split into 10 different spatial reasoning tasks, covering the four DISE quadrants. Secondly, it includes a set of over 12,000 verified 3D spatial reasoning VQA pairs that are generated through an automated pipeline. The synthetic VQA pairs spread across five 3D Spatial Reasoning tasks.

We conducted a comprehensive evaluation across 32 state-of-the-art (SOTA) VLMs on Spatial-DISE. These encompassed a range of advanced VLMs, featuring both proprietary and open-source models: 22 foundation models, 7 reasoning models, and 3 models post-trained with spatial-related datasets. Our findings reveal a profound and universal weakness in current VLMs. Overall performance remains low, with most models scoring only slightly above random chance and far below the human baseline. Our in-depth error analysis further reveals that these failures stem not from simple visual perception, but from fundamental deficits in cognitive processes like rule-based reasoning and mental simulation. Our key contributions include:

- **A Cognitively Grounded Taxonomy:** We adopt a cognitively grounded framework that, unlike previous task-oriented benchmarks, provides a unified taxonomy to classify any spatial task, revealing specific weaknesses like dynamic reasoning that are otherwise obscured.
- **A Scalable and Verifiable Data Generation Pipeline:** We design and implement a novel, automated pipeline using Blender to programmatically generate complex 3D spatial reasoning tasks. This methodology is a key contribution, offering a reusable tool for the community to overcome the data scarcity that has limited previous dynamic reasoning research. The pipeline ensures verifiability through seeded randomization and reproducible distractor generation strategies.
- **A Unified & Verifiable Cognitive Benchmark:** Leveraging our pipeline, we introduce the Spatial-DISE and the accompanying 12,000-VQA dataset, as a benchmark to systematically

¹<https://www.blender.org/>

and extensively evaluate complex cognitive spatial reasoning tasks at a scale sufficient for robust evaluation and future model training.

- **Exploring the Boundaries of Cognitive Spatial Reasoning in VLMs:** By benchmarking 32 SOTA models, we define the current boundaries of VLM capabilities in cognitive spatial reasoning. Our analysis reveals a universal performance ceiling, especially for multi-step mental simulation, highlighting the significant gap between AI and human-level spatial intelligence.

2 RELATED WORK

Table 1: Comparison of Existing Benchmarks under DISE Taxonomy. Abbreviations— I-S: Intrinsic-Static; I-D: Intrinsic-Dynamic; E-S: Extrinsic-Static; E-D: Extrinsic-Dynamic.

Benchmark	Data Scale	Domain	Source	I-S	I-D	E-S	E-D
SpatialRGPT Cheng et al., (2024)	1k+	General	Real-World	✗	✗	✓	✗
BLINK (Fu et al., 2024)	7k+	General	Real-World	✗	✗	✓	✓
VSR (Liu et al., 2023)	10k	General	Real-World	✓	✗	✓	✗
What’s Up (Kamath et al., 2023)	820	General	Real-World	✗	✗	✓	✗
CV-Bench (Tong et al., 2024)	2638	General	Real-World	✗	✗	✓	✗
LEGO-Puzzles (Tang et al., 2025)	1100	Objects	Syn.	✗	✓	✓	✓
COMFORT (Zhang et al., 2025)	1220	Objects	Syn.	✓	✗	✓	✗
3DSRBench (Ma et al., 2025)	2772	General	Real-World	✗	✗	✓	✗
VSI-Bench (Yang et al., 2024)	5k	General	Real-World	✗	✗	✗	✓
Spatial457 (Wang et al., 2025)	20k+	Objects	Syn.	✓	✗	✓	✓
Q-SpatialBench (Liao et al., 2024)	271	General	Real-World	✗	✗	✓	✗
SAT (Ray et al., 2024)	175k	General	Real-World+Syn.	✗	✗	✓	✓
SPARE3D (Han et al., 2020)	10k+	Cognition	Syn.	✓	✗	✗	✗
SpatialEval (Wang et al., 2024)	13k+	Cognition	Real-World	✓	✗	✓	✓
BSA (Xu et al., 2025)	312	Cognition	Real-World	✓	✓	✓	✓
SPACE (Ramakrishnan et al., 2024)	5k+	Cognition	Real-World	✓	✓	✗	✓
OmniSpatial (Jia et al., 2025)	1.5k	General+Cognition	Real-World	✓	✓	✓	✓
Spatial-DISE Bench	559	Cognition	Real-World+Syn.	✓	✓	✓	✓
Spatial-DISE-12K	12k+	Cognition	Real-World+Syn.	✓	✓	✓	✓

The evaluation of spatial reasoning ability in VLMs has been an active area of research, but prior work suffers from critical gaps in scope, cognitive depth, and scale. Table 1 compares existing benchmarks in coverage scope, number of instances, and data sources.

Previous benchmarks offer a fragmented evaluation, lacking a unified cognitive framework. Benchmarks such as LEGO-Puzzles (Tang et al., 2025), SAT (Ray et al., 2024) and VSI-Bench (Yang et al., 2024) are confined to narrow, specific tasks, preventing a holistic assessment of a model’s true spatial abilities. Spatial-DISE overcomes this by introducing a unified 2x2 cognitive taxonomy. This framework, rooted in cognitive science, enables a comprehensive and balanced evaluation, allowing for the precise diagnosis of model weaknesses.

Furthermore, prior benchmarks have a disproportionate focus on static reasoning. A vast number of benchmarks—including SpatialRGPT (Cheng et al., 2024), SPARE3D (Han et al., 2020), VSR (Liu et al., 2023), CV-Bench (Tong et al., 2024), BLINK (Fu et al., 2024), and What’sUp (Kamath et al., 2023), SpatialEval (Wang et al., 2024) primarily test a model’s ability to perceive fixed scenes and relationships. They evaluate what models "see" but not how they can "reason" about potential changes. Spatial-DISE targets this gap by focusing on intricate dynamic reasoning to thoroughly assess cognitive tasks such as 3D rotation and folding.

Finally, while SAT (Ray et al., 2024), SPACE (Ramakrishnan et al., 2024), BSA (Xu et al., 2025) and OmniSpatial (Jia et al., 2025) have begun to explore the dynamic domain, Spatial-DISE’s uniqueness lies in its integration of a cognitively unified framework and a verifiable generation process. Our work complements these existing efforts by providing a structured and reproducible approach to understanding model failures. Although one might consider aggregating existing benchmarks into a meta-benchmark, such a union would remain DISE-imbalanced, format-heterogeneous, and unable to provide verifiable training data for the underserved dynamic quadrants—gaps that Spatial-DISE is specifically designed to address.